Euclid's Elements — Book I
Proposition 20

Formal Reconstruction — Technical Documentation

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The triangle inequality

1 Introduction

Euclid's Proposition I.20 is the triangle inequality in purely synthetic form. For a nondegenerate triangle ABC, Euclid proves that any two sides taken together are greater than the third. For one of the three cyclic cases the statement is

$$BA + AC > BC.$$

The Hilbert reconstruction does not introduce a numerical length function and does not define an arithmetic sum of real numbers. Instead, the sum $BA + AC$ is represented geometrically. Extend BA beyond A to a point D such that

$$B - A - D \qquad \text{and} \qquad AD \cong AC.$$

Then the whole segment BD represents the concatenation of BA with a copy of AC. The required inequality becomes

$$BC < BD.$$

Thus the formal theorem asserts the existence of such a point D together with the strict segment comparison $BC < BD$.

The completed proof remains entirely inside neutral Hilbert geometry. The classical proof uses Proposition I.5, while the current Lean proof calls the generic theorem `hilbert_isosceles_base_angles` directly. Its decisive proposition-level dependency is Proposition I.19. No parallel postulate, numerical angle measure, or numerical segment length is used.

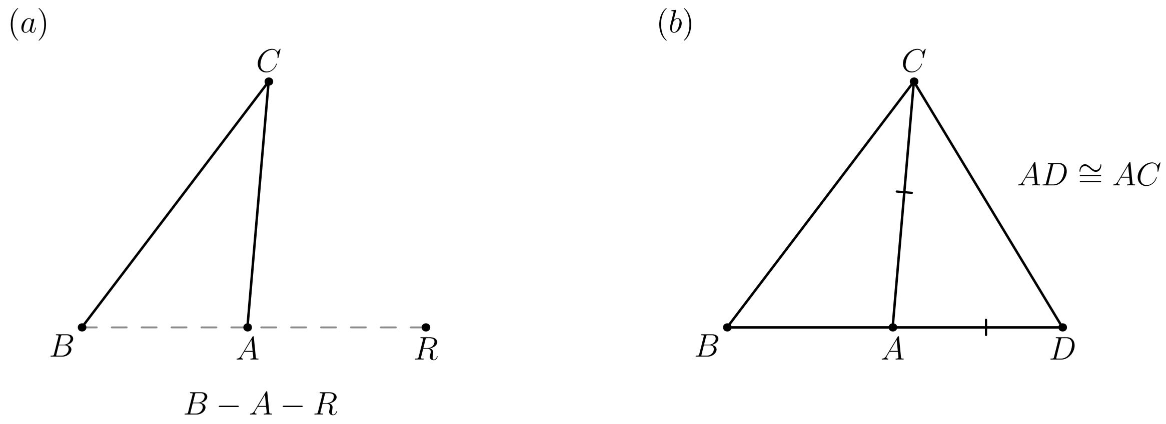

Figure 1: Construction and final configuration for Proposition I.20. (a) The current Lean proof first obtains a hidden extension witness R with $B - A - R$. (b) A copy of AC is then laid off on the ray AR, giving the final point D with $B - A - D$ and $AD \cong AC$. The two panels are deliberately separate because the proof does not require an explicit order relation between R and D.

2 The Classical Configuration

Let ABC be a nondegenerate triangle. Extend BA beyond A to a point D and choose D so that

$$AD \cong AC.$$

The construction records

$$B - A - D, \qquad AD \cong AC.$$

The target is

$$BC < BD.$$

Geometrically, this is exactly the inequality

$$BC < BA + AC.$$

3 Euclid's Classical Proof

After extending BA to D so that $AD = AC$, join CD.

The auxiliary triangle ADC is isosceles. Proposition I.5 therefore gives

$$\angle ADC = \angle ACD.$$

Since A lies between B and D, the ray CA lies inside the angle BCD. Hence

$$\angle ACD < \angle BCD.$$

Combining this with the isosceles-triangle equality gives

$$\angle BDC < \angle BCD.$$

Now apply Proposition I.19 to triangle BCD. The greater angle is opposite the greater side, so

$$BC < BD.$$

Finally, because $B - A - D$ and $AD = AC$, the segment BD is the geometric concatenation of BA and AC. Therefore

$$BA + AC > BC.$$

The classical dependency pattern is

$$\begin{array}{c} B - A - D,\ AD \cong AC \\ \downarrow \\ \angle ADC \cong \angle ACD \quad \text{(I.5)} \\ \downarrow \\ \angle BDC < \angle BCD \\ \downarrow \quad \text{(I.19)} \\ BC < BD. \end{array}$$

4 Synthetic Form of the Triangle Inequality

The formal reconstruction deliberately avoids introducing an arithmetic addition operation on segments. The public result is expressed by an extension point D:

$$\exists D\ (B - A - D\ \wedge\ AD \cong AC\ \wedge\ BC < BD).$$

This is the natural Hilbert form of Euclid's phrase "two sides taken together are greater than the remaining one."

The representation has two advantages. First, it remains purely geometric. Second, it makes explicit which part of the statement belongs to order and congruence geometry: the "sum" is a concatenated segment, not an externally imposed numerical quantity.

5 Proof Architecture of the Reconstruction

The current production module imports only

```
import CGJteamLab.Proposition19
```

and contains one proposition-specific theorem,

```
euclid_proposition_20
```

under

```
[HilbertIncidence Geo]
[HilbertCongruence Geo].
```

It is a direct proof rather than a wrapper around a proposition-local helper.

The current formal chain is

$$
\begin{gathered}
ABC \text{ noncollinear} \\
\downarrow \\
\texttt{between\_extension} \\
\downarrow \\
B - A - R \\
\downarrow \\
\texttt{segment\_construction on ray } AR \\
\downarrow \\
AD \cong AC, \quad \mathrm{SameRay}(A; R, D) \\
\downarrow \\
\texttt{between\_transport\_sameRays} \\
\downarrow \\
B - A - D \\
\downarrow \\
A \text{ witnesses } \mathrm{RayMeetsSegment}(CA, DB) \\
\downarrow \\
\texttt{hilbert\_angleLess\_intro} \\
\downarrow \\
\angle ACD < \angle DCB \\
\downarrow \\
\texttt{hilbert\_isosceles\_base\_angles} \\
\downarrow \\
\angle ADC \cong \angle ACD \\
\downarrow \\
\text{strict-angle transport and same-ray normalization} \\
\downarrow \\
\angle BDC < \angle BCD \\
\downarrow \quad \mathrm{I.19} \\
BC < BD.
\end{gathered}
$$

A crucial audit correction is that the current proof does *not* invoke Proposition I.16. The strict subangle relation is introduced directly from the interior-ray witness by `hilbert_angleLess_intro`. Likewise, the formal proof does not call a proposition-level I.5 theorem; it uses the reusable isosceles-base-angle theorem directly.

6 Diagrammatic Audit

The preserved Book1R chapter had one classical diagram. That final configuration is correct, but it suppresses the genuinely constructed intermediate point R used by the current Lean proof. The audited Figure 1 therefore keeps a single float but uses two panels.

Panel (a) records the hidden extension state $B - A - R$. Panel (b) records the final Euclid/Lean state $B - A - D$, $AD \cong AC$, with CD joined. The two points R and D are not placed on one common panel: from the actual theorem state we need only that D lies on the same ray from A as R, followed by transport of betweenness. Drawing a specific order such as $A - R - D$ or $A - D - R$ would add geometric information not required by the proof.

No separate figure is introduced for the witness A in `HilbertRayMeetsSegment`, for angle congruence transport, for same-ray normalization, or for the final I.19 application. These steps change the logical interpretation of the same final configuration but do not create a new geometric state.

7 Construction of the Extension Point

The point D is constructed in two stages.

First, Hilbert's order axiom extends the line beyond A:

$$B - A - R.$$

In Lean this is obtained from

`HilbertOrder.between_extension`.

Second, a copy of AC is laid off from A on the ray AR using

`HilbertCongruence.segment_construction`.

This produces a point D satisfying

$$AD \cong AC$$

and placing D on the same ray from A as R.

The theorem

`hilbert_between_transport_sameRays`

then transports the original betweenness relation $B - A - R$ to

$$B - A - D.$$

Thus the geometric "sum" segment BD is obtained entirely from existing order and congruence infrastructure.

8 The Interior-Angle Step

The key order-theoretic observation is that the ray CA meets the open segment DB at the point A itself. Since

$$D - A - B,$$

the witness for

`HilbertRayMeetsSegment Geo C A D B`

is simply A.

The strict angle comparison is then introduced directly with `hilbert_angleLess_intro`.

The first angle is $\angle ACD$. The corresponding proper subangle of $\angle DCB$ is represented by $\angle DCA$, which is the same unoriented angle as $\angle ACD$. The representation-level symmetry of angles is supplied by

`angle_congruent_reverse_second`

together with reflexivity of angle congruence.

This yields

$$\angle ACD < \angle DCB.$$

No auxiliary half-plane or perpendicular construction is required. The entire strict comparison follows from the ray meeting the opposite segment.

9 Transport Through the Isosceles Triangle

From

$$AD \cong AC$$

and the noncollinearity of A, D, C, Proposition I.5 is available in the form

`hilbert_isosceles_base_angles`.

It gives

$$\angle ADC \cong \angle ACD.$$

For Proposition I.20 we also introduced the reusable theorem

`hilbert_angleLess_transport_left`.

It expresses invariance of strict angle comparison under replacing the smaller angle by a congruent angle.

Applied here, it transports

$$\angle ACD < \angle DCB$$

across

$$\angle ADC \cong \angle ACD,$$

producing

$$\angle ADC < \angle DCB.$$

Since $B - A - D$, the rays DA and DB coincide. The theorem

`hilbert_angle_eq_of_sameRay_first`

therefore identifies the angle $\angle ADC$ with $\angle BDC$. Hence

$$\angle BDC < \angle DCB.$$

Finally, reversal of the two boundary rays at C identifies

$$\angle DCB \cong \angle BCD.$$

The existing theorem

`hilbert_angleLess_transport_right`

then gives

$$\angle BDC < \angle BCD.$$

10 Application of Proposition I.19

At this point the proof has produced exactly the hypothesis needed for Proposition I.19 in triangle BCD:

$$\angle BDC < \angle BCD.$$

Applying

`euclid_proposition_19`

yields

$$BC < BD.$$

This is the synthetic triangle inequality for the chosen pair of sides. The other two cyclic cases follow by renaming the vertices.

11 Lean Formalization

The public theorem is:

```
theorem euclid_proposition_20
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C : Geo.Point)
    (hABC : not PrimCollinear Geo A B C) :
    exists D : Geo.Point,
      Geo.Between B A D /\
      Geo.Congruent A D A C /\
      HilbertSegmentLess Geo B C B D := by
```

The first stage derives $A \neq B$ and extends BA beyond A:

```
  have hAB : A != B :=
    hilbert_noncollinear_ne_first
      Geo A B C hABC
```

```
have hBA : B != A :=
  hAB.symm

rcases
    HilbertOrder.between_extension
      B A hBA with
  <R, hBAR>
```

A copy of AC is then laid off on the ray AR:

```
have hAR : A != R :=
  (HilbertOrder.between_incidence
    B A R hBAR).2.1

rcases
    HilbertCongruence.segment_construction
      (Geo := Geo)
      A C
      A R
      hAR with
  <D, hRayARD, hAD_AC>
```

Betweenness is transported to the actual constructed point D:

```
have hRayABA :
    HilbertSameRay Geo A B B :=
  hilbert_sameRay_refl
    Geo A B hBA

have hBAD :
    Geo.Between B A D :=
  hilbert_between_transport_sameRays
    Geo
    B A R
    B D
    hBAR
    hRayABA
    hRayARD
```

The theorem then packages the construction and continues with the strict segment comparison:

```
refine <D, hBAD, hAD_AC, ?_>
```

The interior ray is represented directly by the point A:

```
have hDAB :
    Geo.Between D A B :=
  (HilbertOrder.between_incidence
    B A D hBAD).2.2.2.2

have hRayCAA :
    HilbertSameRay Geo C A A :=
  hilbert_sameRay_refl
    Geo C A hAC
```

```
have hInside :
    HilbertRayMeetsSegment Geo C A D B :=
  <A, hDAB, hRayCAA>
```

Angle reversal is obtained through the representation-level congruence laws:

```
have hAngleRefl :
    Geo.AngleCongruent A C D A C D :=
  Geometry.Geo.angle_congruent_reflexive
    Geo A C D

have hAngleSwap :
    Geo.AngleCongruent A C D D C A :=
  (Geo.angle_congruent_reverse_second
    A C D
    A C D).mp
    hAngleRefl
```

This gives the strict subangle relation:

```
have hLessACD_DCB :
    HilbertAngleLess Geo A C D D C B :=
  hilbert_angleLess_intro
    Geo
    A C D
    D C B
    A
    hACD
    hDCB
    hInside
    hAngleSwap
```

Proposition I.5 and left transport give the corresponding comparison with the base angle at D:

```
have hIso :
    Geo.AngleCongruent A D C A C D :=
  hilbert_isosceles_base_angles
    Geo
    A D C
    hADC
    hAD_AC

have hLessADC_DCB :
    HilbertAngleLess Geo A D C D C B :=
  hilbert_angleLess_transport_left
    Geo
    A C D
    A D C
    D C B
    hLessACD_DCB
    hADC
    hIso
```

The ray DA is then replaced by the same ray DB:

```
have hRayDAB :
```

```
    HilbertSameRay Geo D A B :=
  hilbert_sameRay_of_between
    Geo D A B hDAB

have hRayDBA :
    HilbertSameRay Geo D B A :=
  hilbert_sameRay_symm
    Geo D A B hRayDAB

have hBDC_ACD :
    Geo.AngleCongruent B D C A C D := by
  unfold Geometry.Geo.AngleCongruent at hIso |-
  rw [hilbert_angle_eq_of_sameRay_first
    Geo D B A C hRayDBA]
  exact hIso
```

Using left transport again gives

```
have hLessBDC_DCB :
    HilbertAngleLess Geo B D C D C B :=
  hilbert_angleLess_transport_left
    Geo
    A C D
    B D C
    D C B
    hLessACD_DCB
    hBDC
    hBDC_ACD
```

Finally, the right-hand angle is reversed and Proposition I.19 closes the proof:

```
have hAngleReflDCB :
    Geo.AngleCongruent D C B D C B :=
  Geometry.Geo.angle_congruent_reflexive
    Geo D C B

have hDCB_BCD :
    Geo.AngleCongruent D C B B C D :=
  (Geo.angle_congruent_reverse_second
    D C B
    D C B).mp
    hAngleReflDCB

have hLessBDC_BCD :
    HilbertAngleLess Geo B D C B C D :=
  hilbert_angleLess_transport_right
    Geo
    B D C
    D C B
    B C D
    hLessBDC_DCB
    hBCD
    hDCB_BCD

exact
  euclid_proposition_19
    Geo
```

```
    B C D
    hBCD
    hLessBDC_BCD
```

12 Relationship with Earlier Propositions

12.1 Classical Proposition I.5 versus the formal isosceles theorem

Euclid's classical proof uses I.5 after constructing the isosceles triangle ADC. The current Lean theorem does not call `euclid_proposition_5`; it calls `hilbert_isosceles_base_angles` directly. Thus I.5 belongs to the classical proof DAG, while the reusable Hilbert theorem is the direct formal dependency at this step.

12.2 Proposition I.16

The preserved Book1R chapter contained a subsection assigning I.16 the strict-angle step. That description does not match the current production proof. No call to `euclid_proposition_16` occurs in `Proposition20.lean`. Instead, the point A itself witnesses that the ray CA meets the open segment DB, and `hilbert_angleLess_intro` yields

$$\angle ACD < \angle DCB.$$

I.16 remains relevant to an ancient alternative proof mentioned in the historical literature, but not to the current formal DAG of I.20.

12.3 Proposition I.19

Proposition I.19 is the direct proposition-level dependency that closes the proof. Once strict angle transport has produced

$$\angle BDC < \angle BCD,$$

`euclid_proposition_19` gives

$$BC < BD.$$

13 New Reusable Infrastructure

The current I.20 proof uses the reusable Hilbert-interface theorem

`hilbert_angleLess_transport_left`.

Its role is symmetric to

`hilbert_angleLess_transport_right`.

Together these two transport theorems express that strict synthetic angle comparison is invariant under congruent replacement of either compared angle. The preserved Book1R chapter records the left-transport theorem as infrastructure developed for this reconstruction; the present audit does not re-run the historical introduction point.

The proof also reuses the existing construction pattern

$$\texttt{between\_extension} \longrightarrow \texttt{segment\_construction} \longrightarrow \texttt{between\_transport\_sameRays},$$

which already occurs elsewhere in the Hilbert development.

14 Dependency Summary

The formal dependency structure of Proposition I.20 can be summarized as

$$\begin{gathered}
\text{Hilbert incidence + order + congruence} \\
\downarrow \\
\text{between extension + segment construction} \\
\downarrow \\
B - A - D, \quad AD \cong AC \\
\downarrow \\
\texttt{hilbert_isosceles_base_angles + HilbertAngleLess} \\
\downarrow \\
\angle BDC < \angle BCD \\
\downarrow \quad \text{I.19} \\
BC < BD \\
\downarrow \\
BA + AC > BC \quad \text{(synthetic interpretation).}
\end{gathered}$$

No Euclidean parallel axiom appears in this chain. Proposition I.20 is therefore a theorem of neutral geometry in the present Hilbert reconstruction.

15 Conclusion

Proposition I.20 is the first place in this sequence where Euclid's language of adding lengths is naturally reconstructed without introducing numerical length. The formal statement replaces the expression $BA + AC$ by an explicit extension segment BD satisfying

$$B - A - D, \qquad AD \cong AC.$$

The rest of the argument is purely synthetic. Classically I.5 supplies the isosceles base-angle equality; in the current Lean proof this is supplied directly by `hilbert_isosceles_base_angles`. The interior-ray witness produces the strict angle inequality through `hilbert_angleLess_intro`, congruence transport normalizes the two angles, and Proposition I.19 converts the angle inequality back into the desired side inequality.

The resulting Lean proof therefore reproduces the classical geometry while exposing its exact logical structure. The preserved Book1R chapter records no new geometric axiom for this proposition. In this audit the production Lean file was read but no new `lake build` or `#print axioms` check was run.